

# Bhadrinathan A

AI Engineer | Full-Stack Application Development

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## Profile Summary

- **AI and Full-Stack Engineer** with 5+ years of software development experience, including an internship, spanning web platforms, real-time services, and AI applications.
- Hands-on with **RAG, LLM APIs, LangChain, LangGraph, and MCP**, supported by **React/Next.js** frontend and **Python/FastAPI** backend engineering.
- Progressed to **Lead Engineer** at BloomSkillTech, owning architecture, deployment workflows, and mentoring 6+ engineers.

## Technical Skills

|                          |                                                                             |
|--------------------------|-----------------------------------------------------------------------------|
| AI Applications          | RAG, LangChain, LangGraph, MCP, LLM APIs, pgvector, Qdrant                  |
| Programming Languages    | Python, JavaScript, TypeScript, Rust                                        |
| Frontend and Mobile      | React, Next.js, React Native                                                |
| Backend and APIs         | FastAPI, SQLAlchemy, Node.js, Express, Axum, REST, WebSockets, SSE, Celery  |
| Data and Messaging       | PostgreSQL, MongoDB, Redis/Valkey, Alembic, MinIO/S3                        |
| Infrastructure and Tools | Docker, Linux, GitHub Actions, CI/CD, Pytest, Git, Nginx/Traefik, WireGuard |

## Work Experience

### Lead Engineer - EdTech Platform | BloomSkillTech, Salem | Jan 2025 - Present

- Architected a **multi-tenant EdTech platform** spanning **React interfaces, FastAPI, PostgreSQL**, and **LLM API integrations**.
- Built **RAG and multi-agent pipelines** using **LangChain, LangGraph, and MCP** for AI application workflows.
- Optimized RAG indexing and vector queries, reducing response latency by **40%** under concurrent workloads.
- Established **CI/CD pipelines, automated tests, and deployment workflows**; mentored **6+ engineers** through code reviews and architecture guidance.

### Senior Software Engineer | BloomSkillTech, Salem | May 2023 - Dec 2024

- Developed a **cloud labs platform** with interactive coding environments, **secure VPN access**, and isolated **Docker workspaces**.
- Built **FastAPI microservices** for container lifecycles, resource management, and real-time sessions.
- Implemented **Redis caching and asynchronous job queues**, improving API throughput by **50%**.

### Software Developer | BloomSkillTech, Salem | Dec 2022 - May 2023

- Built a **law enforcement CRM** with **React UI, Go REST APIs, PostgreSQL, and Docker**; delivered case timelines, full-text record search, and tracking.

### Software Development Intern | Salem City Cyber Crime Police | Aug 2021 - Dec 2022

- Developed a **website incorporating the station's database** and continued **software maintenance and support** until joining BloomSkillTech.
- Received a **Certificate of Appreciation from Salem City Police in December 2021** for website development contributions.

## Open Source and Projects

- [fastapi-querybuilder](#) | Python, FastAPI | [PyPI](#): Optimized query compilation in an open-source package with **18,000+ downloads**.
- [ZeroCode](#) | Rust: Built a **sandboxed execution engine** with integrated caching and performance optimizations; hardened execution against **130+ adversarial test cases**.

## Achievements

- **Active CTF Player | Professional Chess Player**

## Education

- **B.E. Electronics and Communication Engineering** | 2019 - 2023  
Muthayammal Engineering College, Namakkal